David H. Silver
Independent researcher
Poughkeepsie, NY 12603, USA
silverdavi@gmail.com
ORCID: 0000-0002-3071-304X

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Editor-in-Chief
Population Health Metrics
BioMed Central

Dear Editor,

I am submitting the enclosed manuscript, *"Bounds and Estimators for the Civilian-to-Combatant Death Ratio from Aggregation of Conflicting Sources,"* for consideration as a Research Article in *Population Health Metrics*, contributing to the journal's open collection on *The health impacts of war and armed conflict*.

The manuscript develops a partial-identification and robust-Bayesian framework for estimating the civilian-to-combatant share among the dead when sources have known, opposing political incentives. The framework is then applied to the Gaza war 2023–2026, aggregating Palestinian Central Bureau of Statistics demographics, OHCHR identifications, Gaza Ministry of Health records, OCHA totals, IISS/RUSI manpower estimates, and the published Lancet field-survey series. The work is methodologically complementary to the recent Gaza-specific Bayesian study by Gómez-Ugarte et al. (2025) in your journal, which estimated ∼78,000 conflict deaths and a 34–36 year life-expectancy loss. Where their analysis focuses on *total mortality* given uncertainty in reporting and age-sex distribution, mine focuses on the *decomposition* of those deaths into combatants versus civilians, with explicit handling of belligerent contamination. The two are designed to be cited side by side.

The paper's four contributions are (i) a sharp identified set for the combatant share $q = D_M/D$ under bounded differential exposure of civilian adult men; (ii) a precision-weighted aggregation rule for multiple demographic sources with delta-method standard errors; (iii) a *contradiction radius* ρ that scores any disputed claim by the standardised perturbation of independent inputs needed to make it consistent; and (iv) a closed-form bias-robust credible interval under Huber ε-contamination. The framework is symmetric: it rejects the polar narratives "too many combatants" and "zero combatants" against the same feasible set.

The Gaza application finds that the public Israel Defense Forces claim of 17–25k combatants killed has $\rho \gtrsim 20$ standard errors away from the demographic feasibility region. A spatial Bayesian model on the same independently-measured inputs gives $q = 2.5\%$ with 95% credible interval $[1.6\%, 3.8\%]$ and an implied civilian-to-combatant ratio of 43:1 on direct deaths. The naive "uniform random violence" null is rejected at $p \approx 1.7 \times 10^{-16}$, but rejecting it is necessary, not sufficient, to rationalise any specific combatant share. The point estimates align with the demographic Bayesian estimate of Gómez-Ugarte et al. (2025) once their reporting prior is set to ignore underreporting, and with the field-survey violent-death estimates of Spagat et al. (2026) in *Lancet Global Health*. They are roughly an order of magnitude below the IDF figure—consistent with the recent capture–recapture work by Khatib et al. (2025) in *The Lancet*, the methodological critique of Spiegel and Garry (2024) and Wyner (2025), and the long literature on civilian-targeting and one-sided violence (Eck and Hultman, 2007; Kalyvas, 2006).

The manuscript follows BMC structured-abstract style and includes a complete declarations section. All data inputs are from public sources; code and the spatial Bayesian simulator will be deposited in an open repository on acceptance. This work has not been considered or submitted elsewhere, and I declare no competing interests.

About the author. I am an independent researcher with peer-reviewed publications in *Nature*, *PNAS*, *IEEE Trans. Pattern Analysis and Machine Intelligence*, *Human Reproduction*, and *PLOS ONE*, a Microsoft Research PhD Fellowship in Biomathematics, Bioinformatics, and Computational Biology jointly between the Technion and Microsoft Research Cambridge (UK), and authorship of the peer-reviewed monograph *Beyond Popular Science* (Open Book Publishers, Cambridge, 2026; DOI 10.11647/OBP.0526). The present manuscript is conducted independently and is unaffiliated with any of my industry roles.

Possible reviewers with the right combination of conflict-mortality, demographic, and Bayesian-statistics expertise include Diego Alburez-Gutierrez (MPIDR), Ana C. Gómez-Ugarte (CED), Enrique Acosta (MPIDR/CED), Michael Spagat (Royal Holloway), Patrick Ball (HRDAG), Megan Price (HRDAG), and Francesco Checchi (LSHTM).

Sincerely,

David H. Silver